B.TECH CCE REVISION · PART 9

Engineering Fundamentals

Covers the first-year marksheet subjects Engineering Chemistry, Basic Electrical Technology, Basic Electronics, Mechanics of Solids and Basic Mechanical Engineering, with short notes on Engineering Physics where it connects.

The big ideas from first-year engineering, for general literacy: chemistry (with how paints and coatings work), basic electrical, basic electronics, mechanics of solids, and basic mechanical engineering.

Isagi · November 2026 · Written from general knowledge of the field as of late 2026; check tool versions, prices and vendor specifics against current documentation.

How to use this chapter

These subjects are for breadth and curiosity, not for software work, so this part is deliberately light: for each subject, the five to eight ideas that explain the most about the physical world, in plain language, with no problem-solving. Two of them have direct uses outside software. Chemistry explains how paints, primers and waterproofing work, which is the product category of a paints business. Mechanics and mechanical basics explain how buildings crack, why walls get damp, and how machines like tinting and mixing equipment work.

> **How to read it.** One subject per sitting. For each, watch one good explainer video (Section 7 lists channels), then try to explain the "big ideas" list aloud. That is the whole exercise.

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1. Engineering chemistry, and the chemistry of paint
2. Basic electrical technology
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1. Engineering chemistry

Matter is atoms held together by electrons; nearly everything in materials, corrosion, batteries and coatings follows from how those bonds form and break.

The big ideas

1. **Bonding decides properties. Ionic** bonds (electrons transferred: table salt) give hard, brittle, high-melting crystals that conduct when dissolved. **Covalent** bonds (electrons shared: water, plastics, diamond) give anything from gases to the hardest solids. **Metallic** bonds (a sea of shared electrons) give conductivity, shine and ductility. **Weak intermolecular forces** (hydrogen bonds, van der Waals) decide boiling points and why things stick.
2. **Reactions need energy to start, and catalysts lower that barrier.** The **activation energy** is why paint doesn't cure in the can (or cures only slowly), why heat speeds reactions up (roughly doubling the rate for every 10 °C in many cases) and why catalytic converters work.
3. **Thermodynamics says whether a reaction can happen; kinetics says how fast.** Iron rusting is thermodynamically favourable; coatings work by making it kinetically slow.
4. **Electrochemistry: electrons moving between substances.** Batteries, electroplating and corrosion are the same process. **Corrosion** is a small battery: iron gives up electrons at one spot (the anode) in the presence of water and oxygen, forming rust. Salt water conducts better, so coastal areas like Kerala rust faster. Defences: barrier coatings, **galvanising** (zinc corrodes first, sacrificially), **cathodic protection**, stainless alloys.
5. **Water chemistry. Hardness** comes from dissolved calcium and magnesium salts; it causes scale in boilers and pipes and is treated by ion exchange or reverse osmosis. **pH** measures acidity (7 neutral; below acidic, above alkaline). Fresh cement and plaster are strongly **alkaline** (pH 12–13), which matters for coatings (below).
6. **Polymers** are long chains of repeating units. **Thermoplastics** (polythene, PVC, acrylic) soften when heated and can be reshaped. **Thermosets** (epoxy, polyurethane) cross-link into a rigid network that doesn't melt. Most paint films are polymers that form on the wall.
7. **Fuels and combustion.** Burning releases energy stored in chemical bonds; **calorific value** measures how much. Incomplete combustion produces carbon monoxide and soot. Alternative fuels (CNG, ethanol blends, hydrogen) trade energy density, storage difficulty and emissions differently.
8. **Instrumental analysis**: spectroscopy identifies substances by how they absorb or emit light at specific wavelengths. It's how labs check composition and quality, including colour matching.

The chemistry of paint

A paint is a liquid that turns into a solid protective, decorative film. Four ingredient groups:

Component	Job	Examples
Binder (resin)	Forms the film and glues everything to the surface; decides durability	Acrylic, vinyl-acrylic, alkyd, epoxy, polyurethane
Pigment	Colour and hiding power	Titanium dioxide (white, the costliest common ingredient), iron oxides, organic pigments
Extenders (fillers)	Bulk, texture, cost reduction, sheen control	Calcium carbonate, talc, china clay
Solvent / carrier	Keeps it liquid for application, then evaporates	Water (emulsions), mineral spirits (enamels)
Additives	Small amounts, big effects	Thickeners, dispersants, biocides and fungicides, anti-foam, UV stabilisers

How it dries depends on the binder:

- **Water-based emulsions** (most interior and exterior wall paints): tiny polymer particles are dispersed in water. As the water evaporates, particles crowd together and fuse into a continuous film (**coalescence**). Low odour, low VOCs, quick drying, but they need temperatures above a minimum film-forming temperature.
- **Solvent-based alkyds and enamels** (wood, metal): the binder reacts with oxygen from the air and **cross-links** (oxidative curing), which is why enamels take longer and smell. Hard, glossy, durable films.
- **Two-component systems** (epoxy, PU): a resin and a hardener react chemically when mixed. Very tough, chemical-resistant; limited pot life once mixed.

Why the system matters more than the topcoat:

- **Surface preparation and primers**: fresh plaster's alkalinity attacks some binders, causing efflorescence (white salt deposits) and peeling; **alkali-resistant primers** and curing time prevent this. Metal needs rust removal and anti-corrosive primers. **Putty** fills unevenness for a smooth finish.
- **Moisture** is the main enemy of wall paint. Damp walls lead to blistering, peeling, fungal growth. **Waterproofing** coatings (elastomeric, cementitious, polyurethane) stop water ingress; good exterior paints are designed to let water vapour escape while keeping liquid water out (breathability).
- **Sheen** (matt to gloss) depends on the ratio of pigment and extender to binder (**PVC**, pigment volume concentration). More binder means glossier, more washable and more durable; more filler means flatter and cheaper.
- **VOCs** (volatile organic compounds) are solvents released into the air; regulations and health concerns have pushed the industry toward water-based, low-VOC products.
- **Colour** is made by **tinting**: a machine doses concentrated colourants into a white or pastel base by formula. The same colour can look different across base types and sheens, and under different light (**metamerism**).

2. Basic electrical technology

Electricity is the flow of charge, pushed by voltage and resisted by the material. Everything from a phone charger to the national grid follows a few laws.

1. **Voltage, current and resistance.** Voltage (V, volts) is the push; current (I, amperes) is the flow; resistance (R, ohms) opposes it. **Ohm's law: V = I × R.** The water-pipe analogy: pressure, flow rate, pipe narrowness.
2. **Power: P = V × I** (watts). Energy is power over time, billed in **kilowatt-hours** (a 1,000 W heater for one hour = 1 unit). A 1.5-ton AC uses about 1.5 kW; a ceiling fan about 50–75 W; an LED bulb 9 W.
3. **Kirchhoff's laws**: current into a junction equals current out (charge is conserved); voltages around any closed loop sum to zero (energy is conserved). Together with Ohm's law, they solve any simple circuit.
4. **Series vs parallel.** In **series**, the same current flows through everything and one break stops all (old fairy lights). In **parallel**, each branch gets the full voltage and works independently: household wiring is parallel.
5. **AC vs DC.** Batteries and electronics use **DC** (one direction). The grid uses **AC** (direction reverses 50 times a second in India: 230 V, 50 Hz) because **transformers** can step AC voltage up for long-distance transmission (less current, so far less heat loss, since loss = I^2R) and down for homes. Every phone charger converts AC back to low-voltage DC.
6. **Magnetism and motors.** A current creates a magnetic field; a changing magnetic field induces a current (**Faraday's law**). Motors use the first effect to turn electricity into motion; generators use the second to turn motion into electricity. Transformers use it to change voltage.
7. **Three-phase power**: industry and large buildings use three AC supplies offset in time, which deliver smoother, more efficient power to big motors. Homes usually get single-phase.
8. **Safety. Earthing** gives fault current a safe path to ground. **Fuses and MCBs** cut the circuit on overcurrent (protecting wires from overheating). **RCCBs/ELCBs** cut it when current leaks to earth (protecting people from shock). Current through the body, not voltage alone, is what harms; around 30 mA can be dangerous, which is why RCCBs trip at that level.
9. **Power factor**: with motors and other inductive loads, current and voltage fall out of step, so more current flows for the same useful power. Industries are charged for poor power factor and correct it with capacitors.

3. Basic electronics

Electronics controls electricity with semiconductors: switching it, amplifying it, and turning it into information.

1. **Semiconductors** (silicon) conduct somewhere between metals and insulators, and their conductivity can be controlled by adding tiny amounts of impurities (**doping**): **n-type** has extra electrons, **p-type** has "holes" (missing electrons).
2. **The p-n junction and diodes.** Joining p and n types makes a **diode**, which lets current flow one way only. Uses: **rectifiers** turn AC into DC (with capacitors to smooth it) in every power supply. **LEDs** are diodes that emit light; **solar cells** are diodes run in reverse, turning light into current.
3. **Transistors** use a small signal to control a large current, as an **amplifier** (audio, radio) or a **switch**. As switches, billions of them make the logic gates in Part 3. **BJTs** are current-controlled; **MOSFETs** are voltage-controlled and are what chips are built from.
4. **Passive components**: **resistors** limit current; **capacitors** store charge briefly (smoothing power, filtering signals, timing); **inductors** resist changes in current (filters, power converters).
5. **Analog vs digital.** Analog signals vary continuously (a microphone's voltage); digital signals are 0s and 1s. **ADCs** (analogue-to-digital converters) sample analog signals into numbers; **DACs** convert back. Sampling must be at least twice the highest frequency you want to keep (**Nyquist**): CD audio samples at 44.1 kHz to capture up to ~20 kHz of hearing; speech recognition typically uses 16 kHz.
6. **Op-amps** are versatile amplifier building blocks for filtering, amplifying sensor signals and comparing voltages.
7. **Sensors and actuators**: sensors turn physical quantities (temperature, light, pressure, sound) into electrical signals; actuators (motors, relays, speakers) turn electrical signals into action. A microcontroller in between, with an ADC on the input side, is the basic shape of every IoT device and embedded system.
8. **Integrated circuits** pack whole circuits onto one chip; **Moore's law** (the transistor count doubling roughly every two years) has slowed but explains the last 50 years of computing.

4. Mechanics of solids

How materials carry loads, stretch, bend and break. It explains why buildings stand, why walls crack, and why a beam is shaped like an I.

1. **Stress and strain. Stress** is force per unit area (how hard the material is being pushed or pulled internally); **strain** is the fractional change in length (how much it deforms). Units of stress are pascals (N/m^2), usually MPa.
2. **Elasticity and Hooke's law.** Up to a point, stress is proportional to strain and the material springs back: stress = E × strain, where **Young's modulus E** measures stiffness (steel ~200 GPa, concrete ~25–30 GPa, rubber a tiny fraction of a GPa).
3. **The stress-strain curve.** Beyond the **elastic limit**, a ductile material like steel **yields** and deforms permanently, then hardens, then necks and breaks at its **ultimate strength**. Brittle materials (glass, cast iron, concrete in tension) break with little warning. **Ductility** gives warning before failure, which is why reinforced concrete is designed so the steel yields first.
4. **Tension, compression, shear and torsion.** Concrete is strong in compression but weak in tension (about a tenth), so **steel reinforcement** is placed where tension occurs. Shear is sliding force; torsion is twisting (shafts).
5. **Bending.** A loaded beam bends: its top is squeezed, its bottom stretched, and its middle (the **neutral axis**) carries little stress. So material far from the middle does the most work, which is why **I-beams** put most of their steel in the top and bottom flanges. A beam's stiffness rises with the cube of its depth: doubling depth makes it eight times stiffer.
6. **Thermal stress.** Materials expand when heated. If they're restrained, stress builds up. **Expansion joints** in bridges, roads and long walls let them move; without them, things crack. Different materials expanding differently (plaster vs concrete, paint vs substrate) cause cracks and peeling at their boundaries.
7. **Factor of safety**: designs carry several times the expected load to cover uncertainty in loads, materials and workmanship.
8. **Fatigue and creep.** Repeated loading well below the breaking strength can grow tiny cracks until failure (**fatigue**: aircraft, bridges, crankshafts). Under constant load over years, some materials slowly deform (**creep**: concrete, plastics, hot metal).

Why walls crack, in this language: shrinkage as concrete and plaster dry (restrained shrinkage creates tension, and concrete is weak in tension), thermal movement, settlement of foundations, and different materials meeting. Hairline shrinkage cracks are cosmetic and handled with crack fillers and elastomeric paints that stretch; structural cracks (wide, growing, diagonal) need an engineer.

5. Basic mechanical engineering

Turning energy into useful motion, and moving heat where you want it.

1. **Energy conversion and the laws of thermodynamics.** Energy is conserved (**first law**), but every conversion to useful work loses some as waste heat (**second law**). No engine can be 100% efficient; the **Carnot limit** sets the maximum based on the temperature difference it works across.
2. **Heat engines. Internal combustion engines** (petrol, diesel) burn fuel inside cylinders: the four-stroke cycle is intake, compression, power, exhaust. Petrol engines ignite with a spark; diesel engines compress air until it's hot enough to ignite the fuel, which is why they're more efficient. Car engines convert roughly a third of fuel energy to motion; the rest is heat.
3. **Refrigeration and air conditioning** run a heat engine in reverse: a **refrigerant** absorbs heat as it evaporates inside, a **compressor** squeezes it, and it releases heat as it condenses outside. **COP** (coefficient of performance) measures how much heat is moved per unit of electricity; good ACs move 3–5 units of heat per unit of electricity. Inverter ACs vary compressor speed instead of switching on and off, saving energy.
4. **Heat transfer** happens by **conduction** (through solids), **convection** (by moving fluid) and **radiation** (infrared). Insulation, double glazing and reflective "cool roof" coatings work by blocking one or more of these.
5. **Power transmission**: gears, belts, chains and shafts transfer motion and trade speed for torque. A gearbox lets an engine run at efficient speeds across a wide range of vehicle speeds. Power = torque × rotational speed.
6. **Fluid mechanics basics**: pressure in a fluid acts equally in all directions (**Pascal's principle**, used in hydraulic lifts and brakes); faster-moving fluid has lower pressure (**Bernoulli**), which helps explain lift and spray nozzles. **Viscosity** is a fluid's resistance to flow; paint is designed to thin when brushed or sprayed and thicken at rest so it doesn't sag (**shear thinning**).
7. **Manufacturing processes**: casting (pouring molten metal into moulds), forming (forging, rolling), machining (removing material on lathes and mills, now mostly CNC), joining (welding, fastening) and additive manufacturing (3D printing). Choice depends on volume, precision, material and cost.
8. **Machines in a paint business**: high-speed **dispersers** break up pigment clusters in the binder; **bead mills** grind them finer; **tinting machines** dose colourants by volume to a formula; **gyroshakers** mix the can. Accuracy of dosing and dispersion quality decide colour consistency.

6. Self-check questions

1. **Why does iron rust faster near the sea?** Rust is an electrochemical process needing water and oxygen; salt water conducts better, speeding the reaction.
2. **How does galvanising protect steel?** Zinc is more reactive, so it corrodes first (sacrificially) and also forms a barrier.
3. **What are the main parts of a paint, and what does the binder do?** Binder, pigment, extenders, solvent or carrier, additives. The binder forms the film and holds everything to the surface; it decides durability.
4. **How does a water-based emulsion form a film?** Water evaporates, polymer particles pack together and fuse into a continuous film (coalescence).
5. **Why use an alkali-resistant primer on fresh plaster?** Fresh plaster and cement are strongly alkaline and can attack paint binders, causing peeling and efflorescence.
6. **Why is household wiring parallel?** Each appliance gets the full voltage and works independently of the others.
7. **Why is power transmitted at high voltage?** For the same power, higher voltage means lower current, and resistive losses scale with current squared.
8. **MCB vs RCCB?** An MCB trips on overcurrent to protect wiring; an RCCB trips on leakage to earth to protect people.
9. **What does a diode do, and where is it used?** Lets current flow one way; used in rectifiers (AC to DC), LEDs and solar cells.
10. **Why must audio be sampled at more than twice its highest frequency?** The Nyquist limit: below that, higher frequencies can't be captured correctly and get distorted into false lower ones.
11. **Why is concrete reinforced with steel?** Concrete is strong in compression but weak in tension; steel carries the tension.
12. **Why are beams shaped like an I?** Bending stress is greatest far from the neutral axis, so placing material in the flanges gives the most stiffness per kilogram.
13. **Why do walls get hairline cracks?** Shrinkage as plaster and concrete dry, thermal movement, and different materials meeting, all creating tension that these materials resist poorly.
14. **How does an AC cool a room?** Refrigerant absorbs heat as it evaporates inside, is compressed, and releases that heat as it condenses outside.

7. Resources

Check these are still available before committing time.

- **Practical Engineering** (YouTube, Grady Hillhouse). Civil and structural engineering explained with demonstrations; ideal for mechanics of solids.
- **The Efficient Engineer** (YouTube). Clear animated explanations of stress, bending, fatigue, thermodynamics and fluid mechanics.
- **Ben Eater** (YouTube) and **EEVblog** for electronics, from diodes and transistors upwards.
- **Crash Course Chemistry** and **Crash Course Physics** (YouTube) for broad refreshers.
- **Engineering Mindset** (YouTube) for electrical systems, HVAC and motors.
- **Paint and coatings basics**: technical data sheets from major paint manufacturers explain products system by system (primer, putty, topcoat, waterproofing); trade bodies such as the Indian Paint Association publish industry material.